

The 2017 Economist review presents a thoughtful overview of how quantum technologies were transitioning from theoretical physics into early engineering domains, particularly in communications, computation, and national strategic investment. It highlights a global race between governments and corporations to establish leadership, suggesting that disciplines once considered peripheral to physics could become central to future industries.

From a 2026 perspective, this projection has largely materialized, albeit more gradually and systematically than early optimism suggested. Large-scale superconducting systems, such as Google's recent demonstrations using multi-qubit "quantum echo" techniques to stabilize entangled states, indicate progress toward error-corrected logical qubits. In parallel, software layers such as OpenQASM and Microsoft's Q# language, together with intermediate representations like QIR and cloud platforms from IBM, Microsoft, and Amazon, have transformed quantum computing into a collaborative and programmable ecosystem rather than isolated laboratory experiments. Alternative hardware approaches, including Microsoft's research into topological qubits based on Majorana quasiparticles, continue to explore pathways to intrinsically more stable qubits, although these remain under active scientific validation.

From a regional perspective, it is encouraging that quantum development is becoming more globally distributed. South Africa's SA QuTI initiative, alongside research at Stellenbosch University and support from national bodies such as the NRF and CSIR, demonstrates growing capacity in quantum communication and photonics research.

From my professional perspective working in Azure, AI, and DevOps, I have explored how AI can assist in understanding complex quantum problems through my Quantum Grand Challenges project (<https://aka.ms/quantumgrandchallenges>

), illustrating how cloud, AI, and quantum technologies are beginning to converge in practice.

Overall, the Economist article was directionally accurate. While full fault tolerance has not yet been achieved, the convergence of hardware, software, and global collaboration suggests that practical quantum advantage is a realistic medium-term objective.